Self-powered co-harvesting for batteryless sensing and monitoring in two-wheeled IoT applications

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G. Pasquini, M. Doglioni, A. Saviane and D. Brunelli (2025), "Self-Powered Co-Harvesting for Batteryless Sensing and Monitoring in Two-Wheeled IoT Applications," in IEEE STAR 2025.

Cycling is becoming an IoT platform

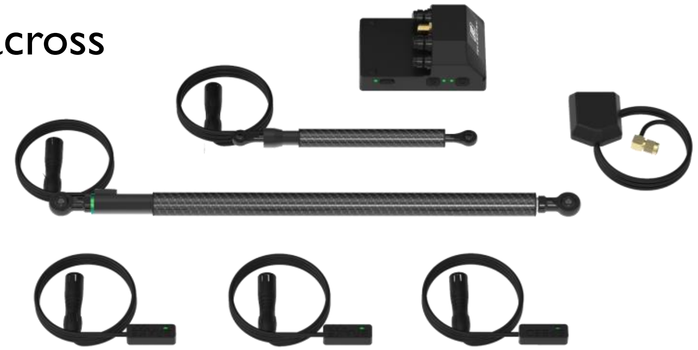
Electronics are now part of modern bikes

Sensors and telemetry

Wireless controls and actuators

This integration delivers tangible improvements across critical functional domains:

- Real-time **performance monitoring**
- Improved **safety enhancement**
- Smarter **bike setup optimization**



BYB telemetry & Fox Transfer Neo Remote controller

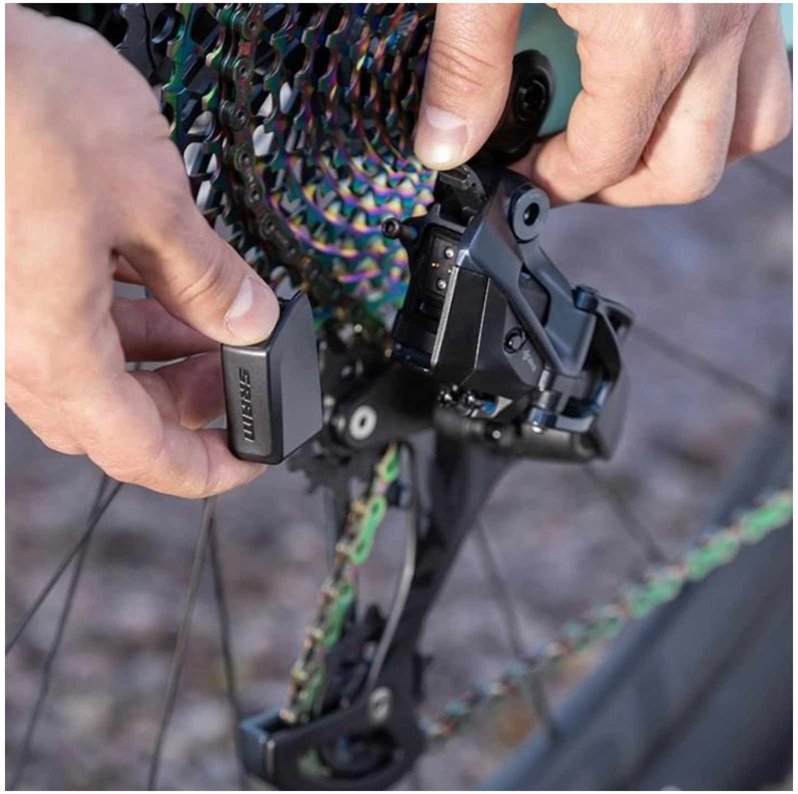
IEEE STAR Trento2025

The problem: too many batteries

Up to 5–7 batteries on a high-end MTB

- Recharging burden and failures
- Environmental impact

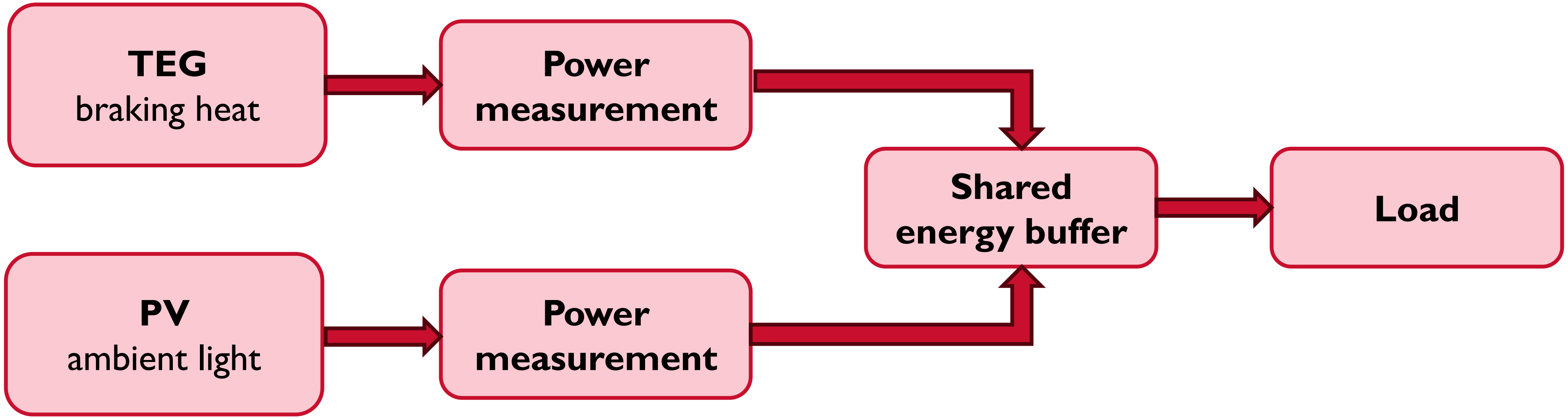
[SRAM Eagle AXS](#)



We need autonomy, not more batteries

Core idea

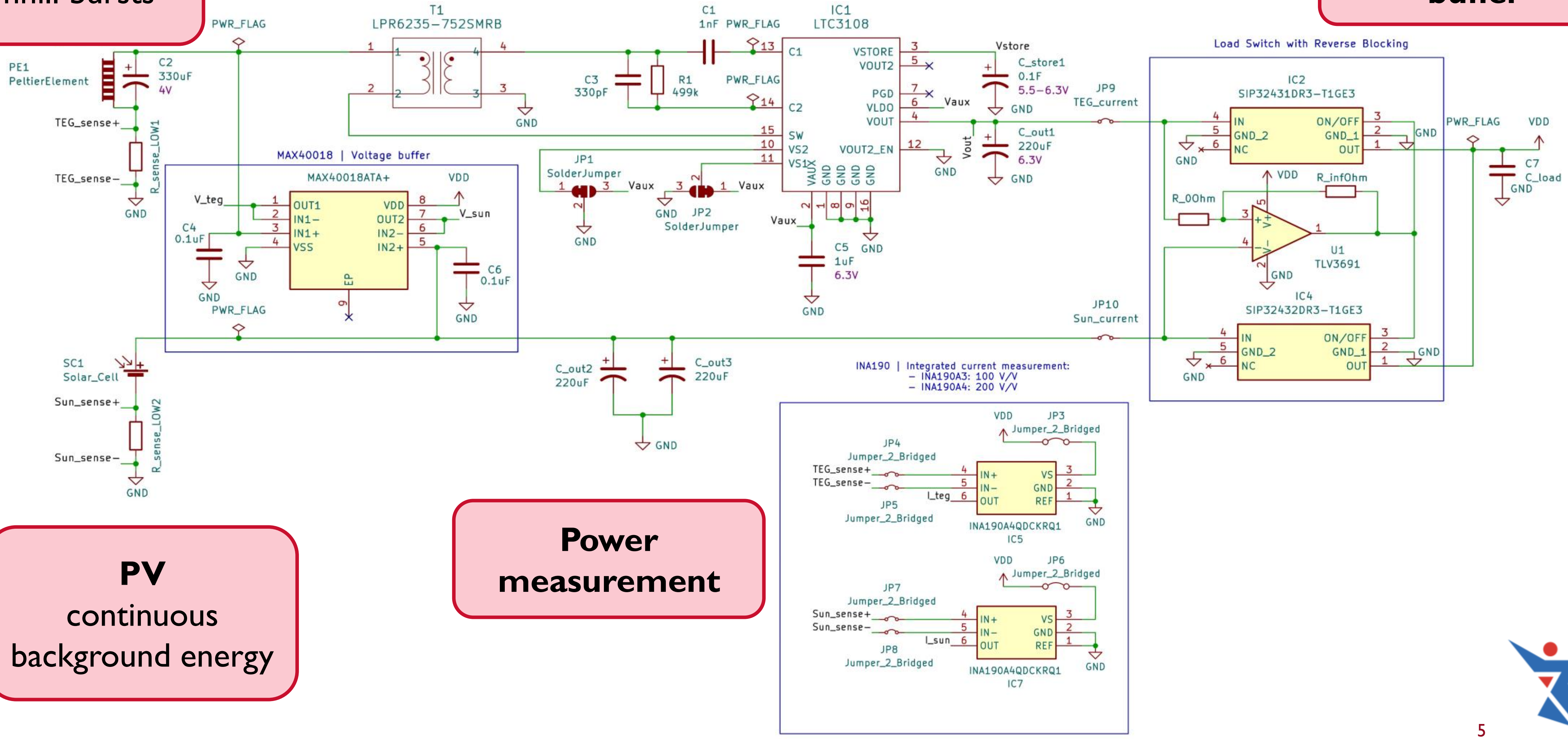
- Hybrid harvesting using **TEG** on brake pads + small **PV cells**
- The two sources are **naturally complementary** in mountain biking
- Provides **friction-free energy** under highly variable conditions
- Harvesting adapts to both braking events (TEG) and ambient light (PV)
- **Power sources also act as sensors** (TEG ↔ brake temperature ↔ braking events)



Hybrid system concept

TEG
downhill bursts

Shared energy buffer



PV
continuous
background energy

Power measurement

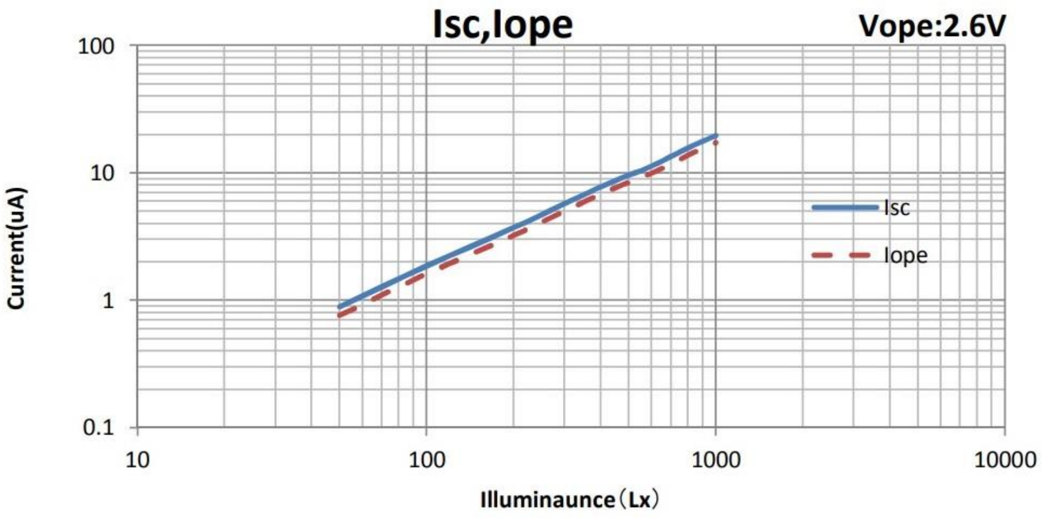
Solar harvesting (PV)

- Specific **PV** for **low-light conditions**, including cloudy or indoor environments
- Generates **µW** to **mW** depending on sunlight
- Provides **baseline power** over the full ride

PV (*AM-1606C-MEL*), mounted on BFTAG01.A sensor tag based on STM32L0

Output power across common lighting conditions:

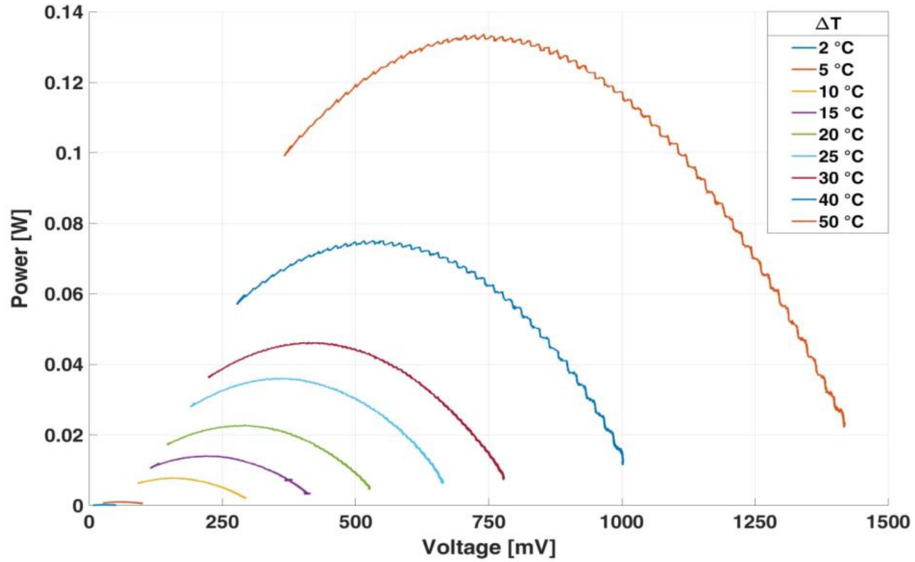
- **Indoor/shade**: ~8–30 µW
- **Daylight**: ~170 µW
- **Direct sunlight**: ~1 mW



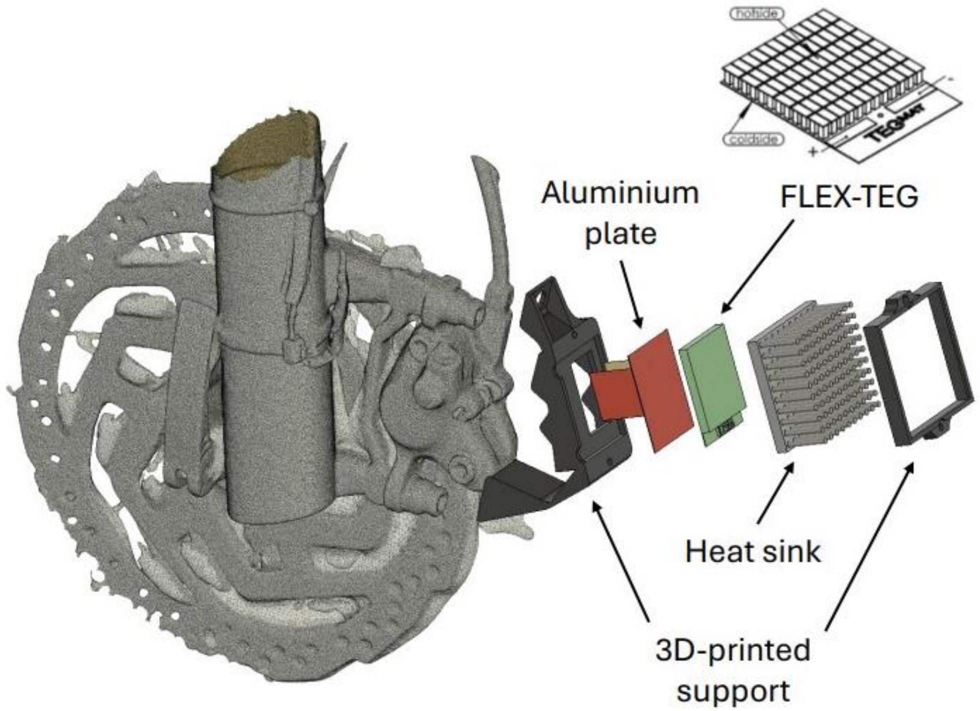
PV Current vs. Illuminance at Vope

Thermoelectric harvesting (TEG)

- Mounted on the brake caliper using the natural **temperature gradient (ΔT)**
- Generates **mW-level bursts** during braking events
- Output directly follows **thermal dynamics**



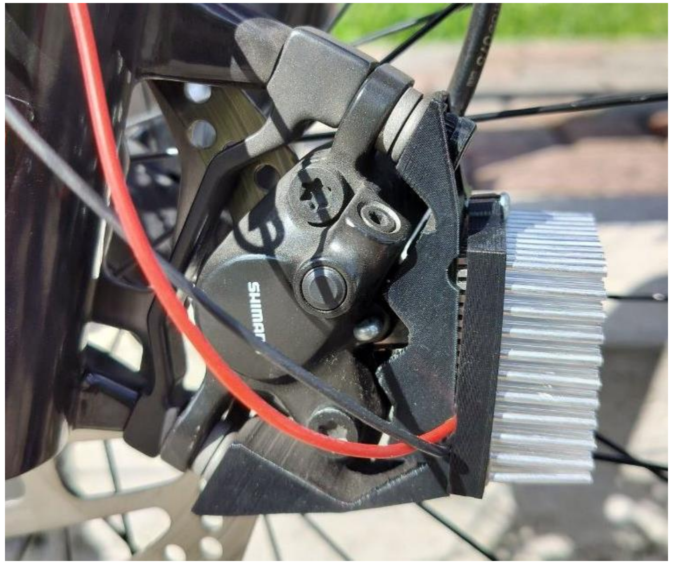
Lab characterization, power generation depending on ΔT.



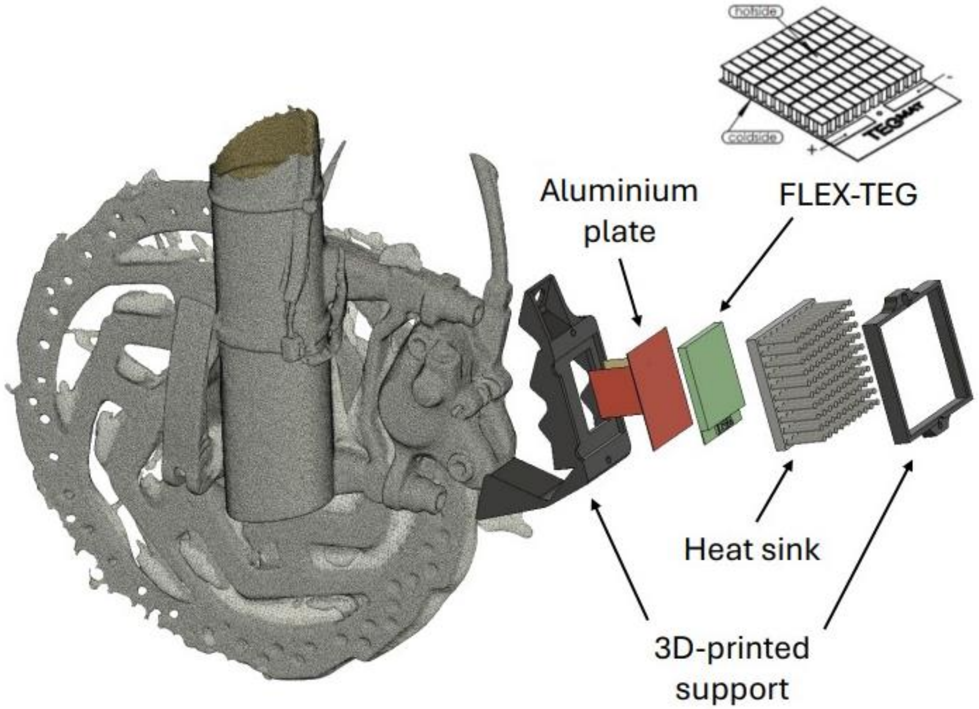
TEG supporting structure

Real TEG setup on the bike

- Custom **3D-printed mount**
- **Aluminum plate** in direct contact with the brake caliper (hot side)
- **Heat sink on the cold** side to maximize ΔT and harvesting efficiency
- TEG mounted **without friction** on the wheel system



TEG setup



TEG supporting structure

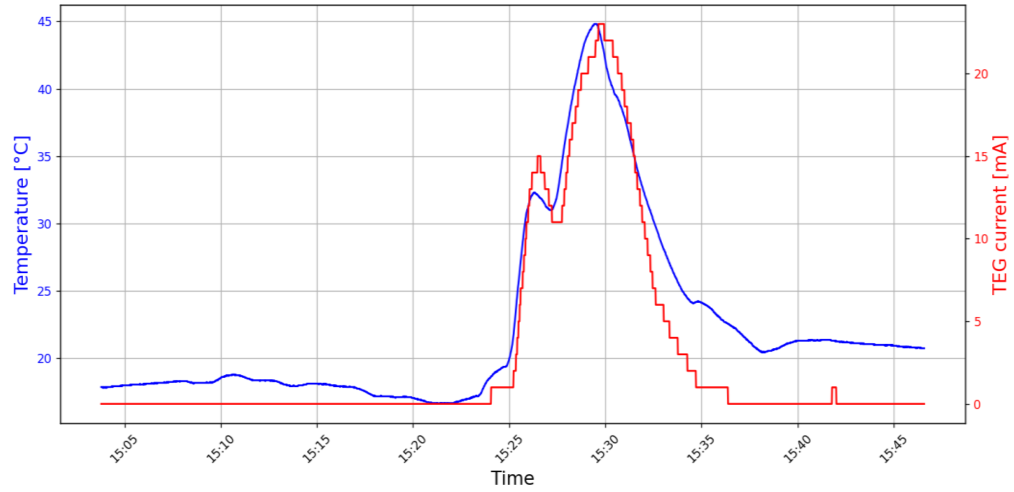
TEG on-field results

Ride 1

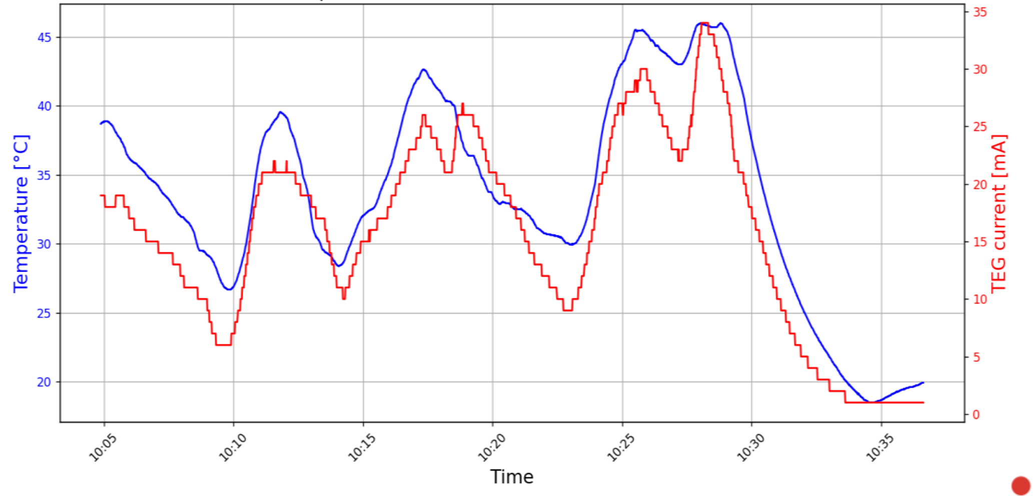
- 1.5 km downhill segment
- Maximum TEG hot-side temperature: 45°C
- Ambient temperature: ~20°C
- **Peak power: 2.3 mW**
- **Harvested energy: 439 J**

Ride 2

- longer descent (~6 km),
- Maximum TEG hot-side temperature: 46°C
- Ambient temperature: ~16°C
- **Peak power: 3.2 mW**
- **Harvested energy: 2236 J**



Hot side temperature and current production over time in RIDE 1

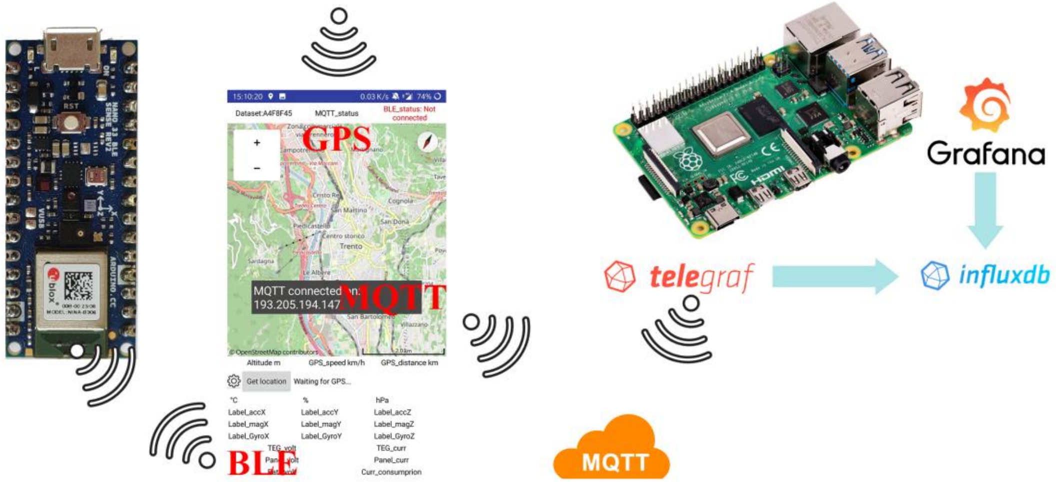


Hot side temperature and current production over time in RIDE 2

What we do with the energy: Two Use Cases

Use Case #1: IoT pipeline for batteryless telemetry

- **Bike setup** & configuration
- **Performance** monitoring
- Track **real-time data** on usage
- **Data visualization** & insights



BLE → Smartphone App → Cloud (MQTT) → Dashboard for Real-time data and visualization

Use Case #2: Power commercial actuators and remote controllers

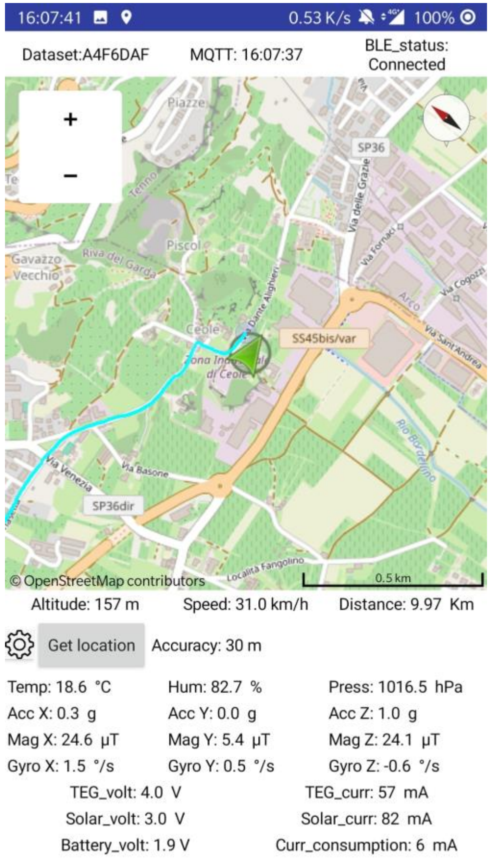
- Electronic shifting systems
- Seatpost droppers

Use Case #1: Batteryless Telemetry

Harvested energy powers a self-sustaining sensor node for sensing, processing and wireless transmission.

Strategy: Energy is buffered on a capacitor and the node wakes only when sufficiently charged

- **Required energy per cycle: 0.133 mJ** Brake temperature, IMU, suspension
- Continuous sampling during downhill (TEG with ΔT > 20 °C)
- 0.77 s sampling period with PV only (10.000 lux)



Use Case #2: Powering Commercial Devices

- **Goal**: Use harvested energy to drive commercial on-board actuators and wireless components (today battery-powered)
- **Works today** (low-power): remotes, shifters, power meters (**1–4 mW**)
- **Not yet** (high-power): droppers, derailleurs, active suspension (**30–300 mW**)

- Device optimization is key: a dropper can work battery-free with short, limited pushes powered by a capacitor, but the trade-off is reduced continuous actuation.
- Continuous actuation: needs optimization.

Fox Transfer Neo Dropper

SRAM Eagle AXS

Key takeaways

Hybrid harvesting

stable & reliable power profile

Results

zero friction, zero maintenance, zero charging

Telemetry

batteryless today (fully supported)

Commercial devices

partially supported (optimization needed)

Conclusions & future work

- **Improve heat transfer and TEG efficiency** to approach lab-scale performance in real riding conditions
- Add more energy sources like **vibration energy harvesting** for rough terrains in MTB use cases
- Extend the architecture to e-bikes and **additional sensors** for telemetry
- Move toward fully batteryless on-bike electronics for maintenance-free IoT systems

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Thank you for your attention

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